Plants on a grid, not on a country average

The question

Module 03 gave every thermal plant its country's projected change in hot days. What does that cost? Join the same plants to the 0.25 degree cells they sit in and compare.

The estimand

The share of 259,319 MW of thermal capacity in eleven countries held by plants whose own grid cell shows days at or above a 35 C heat index rising by more than k=2 days, 1995-2014 to 2040-2059, SSP2-4.5, CMIP6 ensemble median; and its difference from the share obtained by assigning each plant its country's value.

The answer

Grid cells: 17.7% [13.1%, 22.5%] of capacity is exposed. Country values: 42.9% [36.6%, 49.3%]. The difference is -25.2 points [-31.4, -19.4], resampling sites, with the grid and the country measure computed on each resample together. The interval excludes zero. It is not a rounding error between two estimates of one thing: 28.1% of capacity changes class, 26.7% is called exposed by country and not by cell, and only 1.4% the other way.

Method and inputs

Inputs, all fetched by src/fetch.py on 2026-10-19

Plants: WRI Global Power Plant Database (coordinates, MW, fuel).
Hazard: World Bank CCKP CMIP6 0.25 degree annual climatology of hd35 (days at or above 35 C heat index), ensemble median, p10 and p90, from the public S3 bucket wbg-cckp.
Borders: Natural Earth 1:50m. Country values: the CCKP API (validation and comparator).
URL, date, size and SHA-256 of every file are in results/fetch_log.json.

Reading the grid

The files are HDF5; netCDF4 and h5py are outside the allowed libraries, so src/h5lite.py reads the one layout they use. It is checked against the CCKP country API (page 4).

Join

Thermal = coal, oil, petcoke, gas, as in module 03: 676 plants, 260,626 MW.
A plant's hazard is the change in the cell containing it (radius 0), swept to 25, 50 and 100 km means, with and without sea cells. Exposed means strictly more than k days.
Coordinate check: 1 plant had latitude and longitude swapped and is repaired; 4 fall more than 10 km outside their country's European polygon (1,306 MW, 0.50%: Escombreras at latitude 7.6, and Ceuta and Melilla, exclaves outside the mainland polygon) and leave both measures.
The module reproduces module 03's country-level 43.2% on all 676 plants.

Intervals

2,000 bootstrap resamples of 617 sites (units on one coordinate are resampled together). Grid and country measure share each resample. The hazard error is not drawn per plant: ensemble p10 and p90 layers move every plant together (page 5).

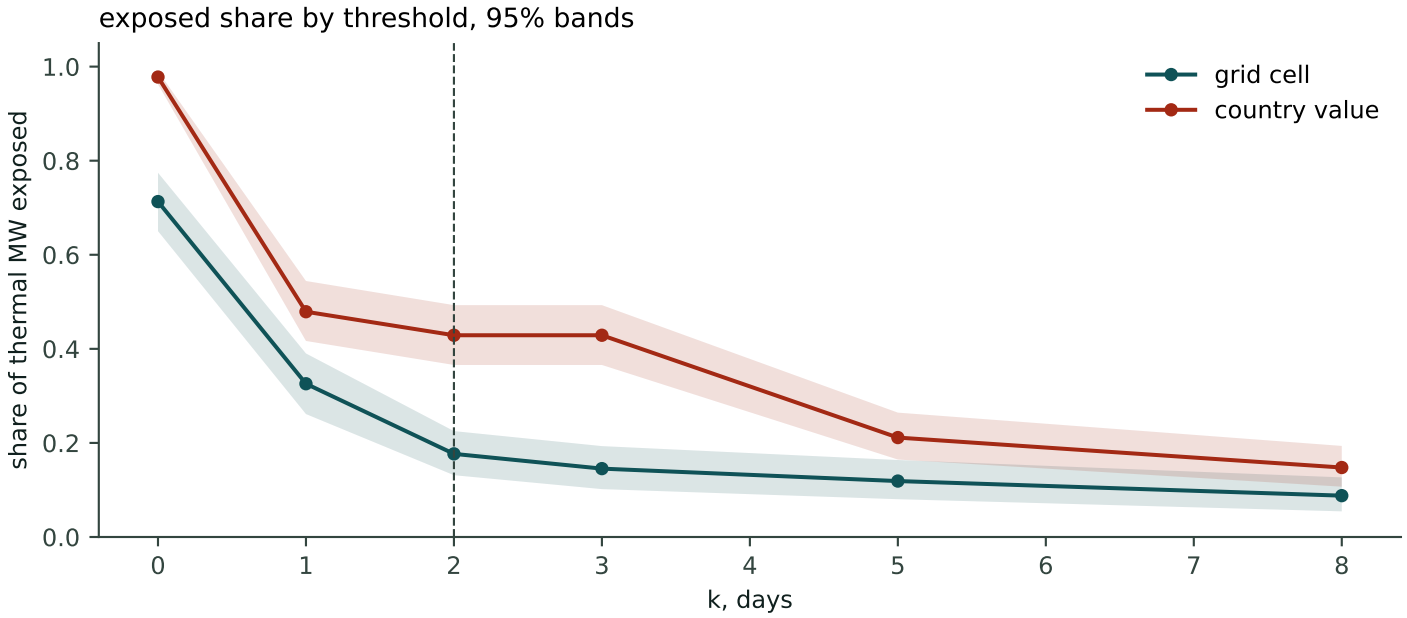
Result: the country value sits above most of its plants

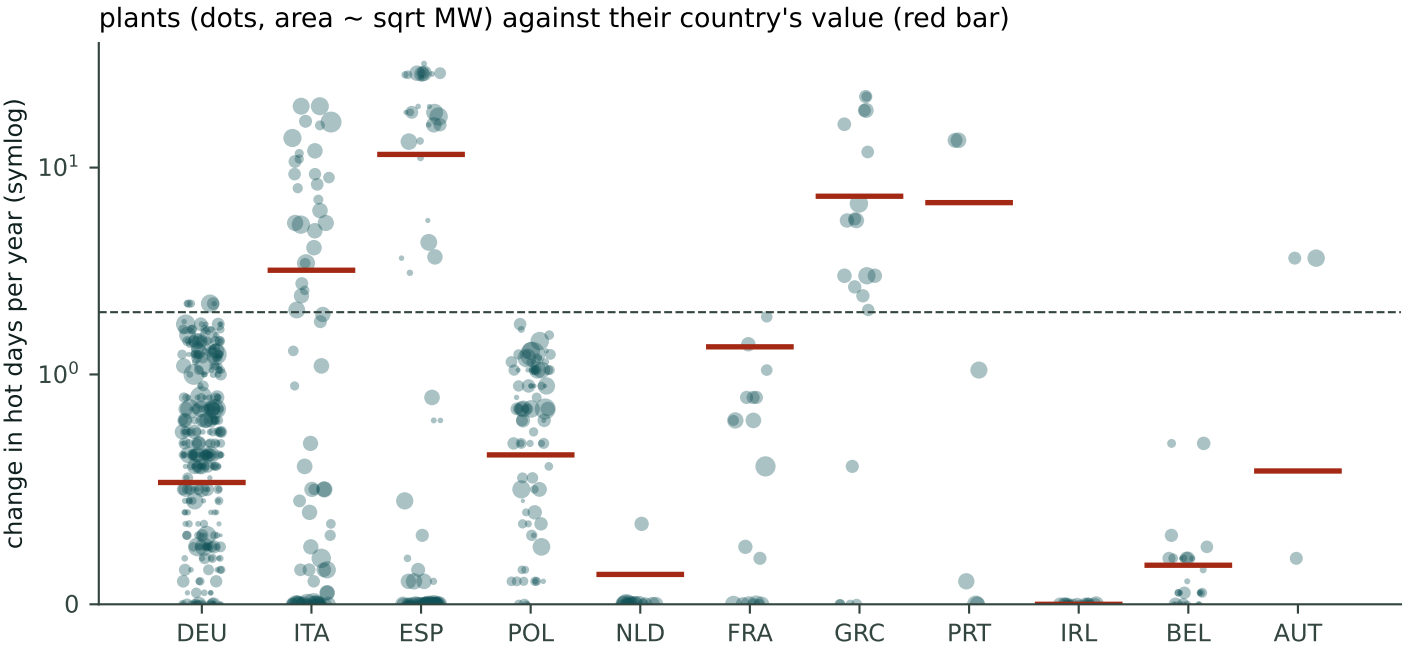
Headline

At k=2: grid 17.7% [13.1%, 22.5%], country 42.9% [36.6%, 49.3%], gap -25.2 points.
The grid point estimate is below the country one at every k from 0 to 8 (from -28.4 to -6.0 points as a range of the gap). Ten plants hold 34.7% of the grid-exposed MW.

Where it comes from

The country figure is an area mean. Spain's is 11.6 days, its plants' MW-weighted mean 5.1.
Country value calls all of Spain, Italy and Portugal exposed; by cell 28%, 36% and 26% of their MW is.
Greece agrees: 90% of its MW is exposed by cell.
Order across the eleven countries survives (Spearman 0.93): the level moves, the ranking barely.





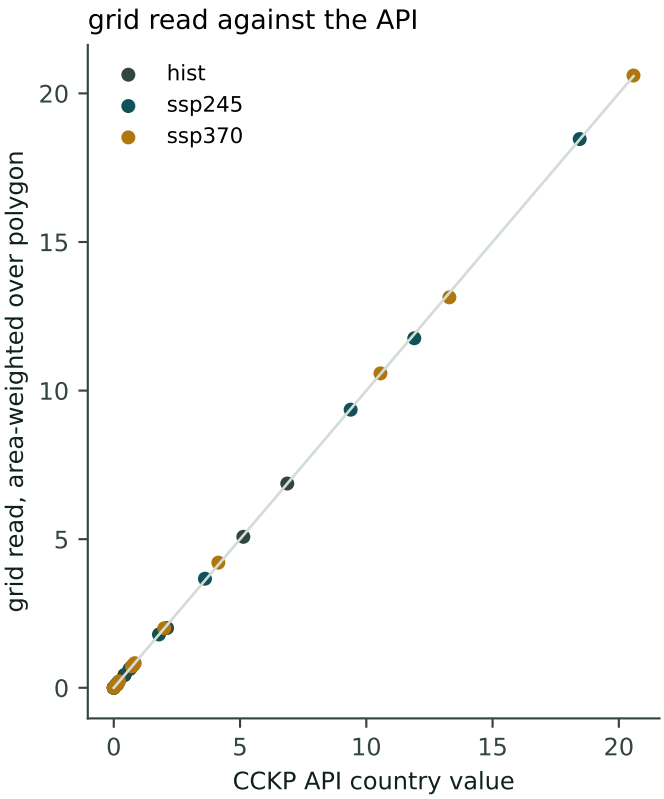
Validation and what resolution buys

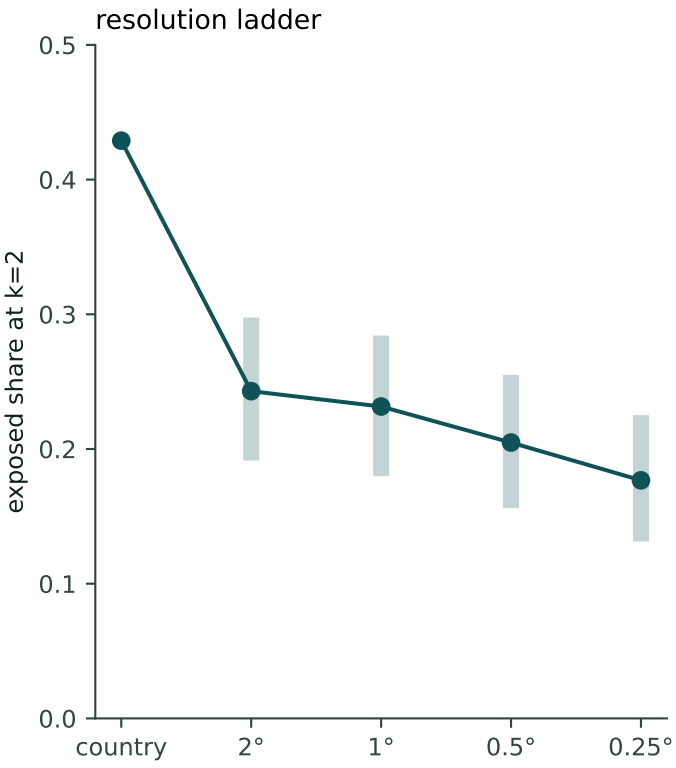
Against something I did not fit

Area-weighted means of the grid over each country's polygon reproduce the CCKP country API for 3 layers in 11 countries; the largest absolute difference is 0.15 days. That checks the HDF5 reader, the grid indexing and the polygon mask together. France matches only when overseas territories are excluded, which is what the API does. It does not validate the model: the same CMIP6 ensemble sits behind both.

Where along the resolution ladder the gap closes

Averaging the cells into blocks shows where the step happens. Country to 2 degree blocks takes 18.6 points off the 25.2 points gap; 2 degrees to 0.25 degrees takes 6.6 points [1.9, 11.4].
Most of the gap is already there at 2 degrees; the 0.25 degree cell adds a smaller step.





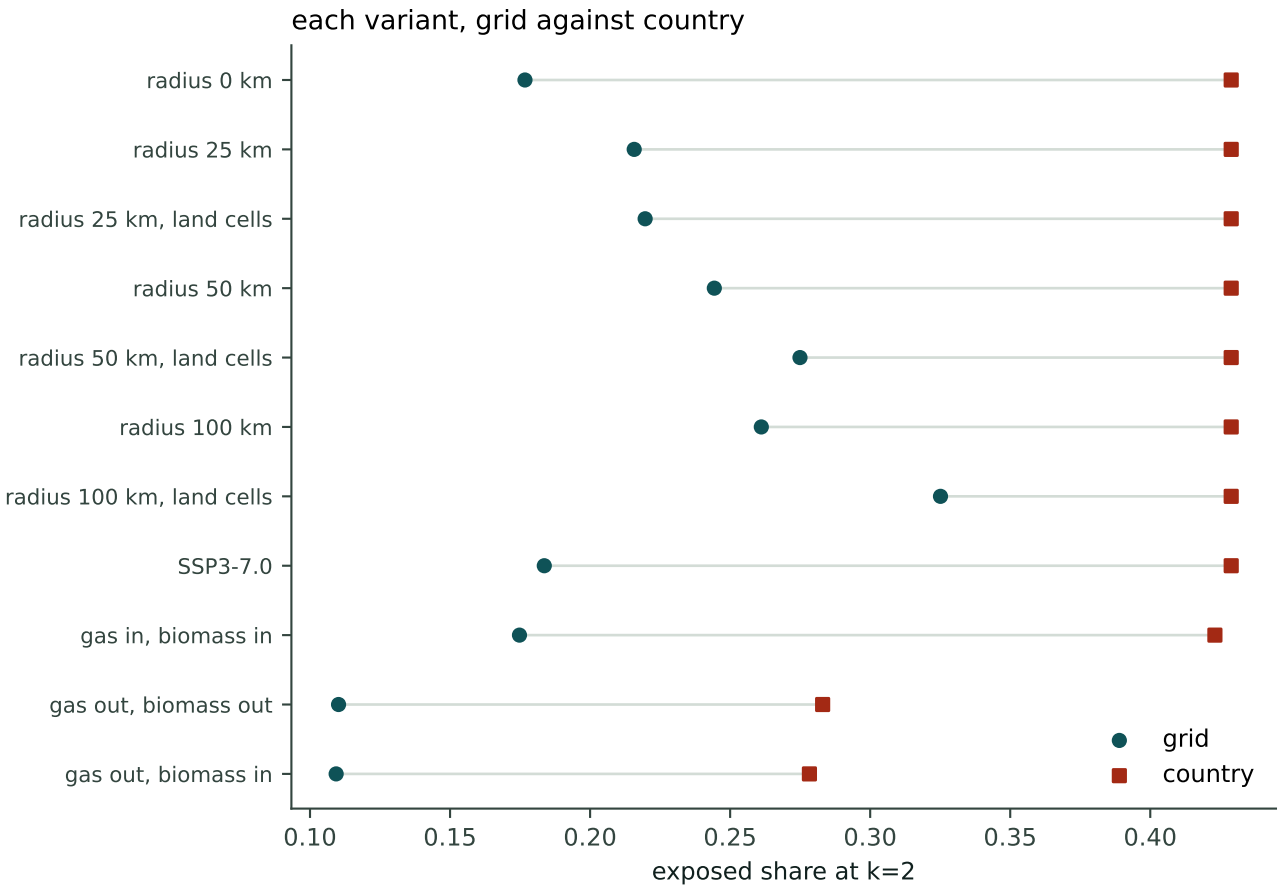
Sensitivity: what moves, and what does not

The arbitrary choices

Matching radius (0 to 100 km, land cells or all): grid share 17.7% to 32.5%, a range of 14.8 points.
Fuel definition: 6.7 points. Scenario, SSP2-4.5 against SSP3-7.0: 0.7 points.
Threshold k=1 against k=3: 18.0 points, more than any of the others; module 03 found the fuel definition larger than that step at country resolution, because a country value moves in jumps. The gap is negative in every variant, from -25.2 to -10.4 points.
Ranking of countries by exposed share: Spearman 0.66 at worst against the baseline.

Hazard uncertainty, drawn once for all plants

The ensemble p10 and p90 layers give a grid share of 10.5% and 46.6%. The country measure moves with them (21.1%, 60.8%). The gap stays negative: -10.7 [-16.1, -5.1] at p10, -14.2 [-22.0, -6.0] at p90. p10 and p90 are separate ensemble layers, not percentiles of the change itself, so they bound the hazard approximately.



Limits, and what I would build next

The assumption that breaks the conclusion

The 0.25 degree files are a regridded product of coarser CMIP6 models; I have not checked CCKP's documentation for their native resolution. If the within-country structure is the models' smooth field and not local conditions at the plant, the grid share is wrong in a direction this module cannot see. The ladder bounds the exposure to this: 6.6 points of the 25.2 points gap lies between 2 degrees and 0.25, so the finest scale is the least secure part of the result. A downscaled or station-based layer could move it either way.

What a sceptic attacks first

1. The 35 C heat-index count is one hazard. A plant's exposure (cooling water, load) is not hd35.
2. The register: WRI has swapped and mistyped coordinates (5 found, only because they leave the country); an error inside the country, or under 10 km, passes the test unseen.
3. The country comparator is the CCKP area mean, the number module 03 used. A national value weighted by population or capacity is not tested here; it would shrink the gap if it weights the hot places less.

No backtest, and why

Nothing here is fitted to a time series. The projection is CCKP's CMIP6 ensemble, consumed unchanged; mid-century has not happened, so there is no realised year to test against. The validation on page 4 checks the read, not the climate.

Next

Replace hd35 with a plant-relevant hazard (cooling-water temperature, extreme-heat derating), and a downscaled layer for the comparison at the plant. Add the ensemble members, not just p10 and p90, so the hazard error has a proper distribution.